

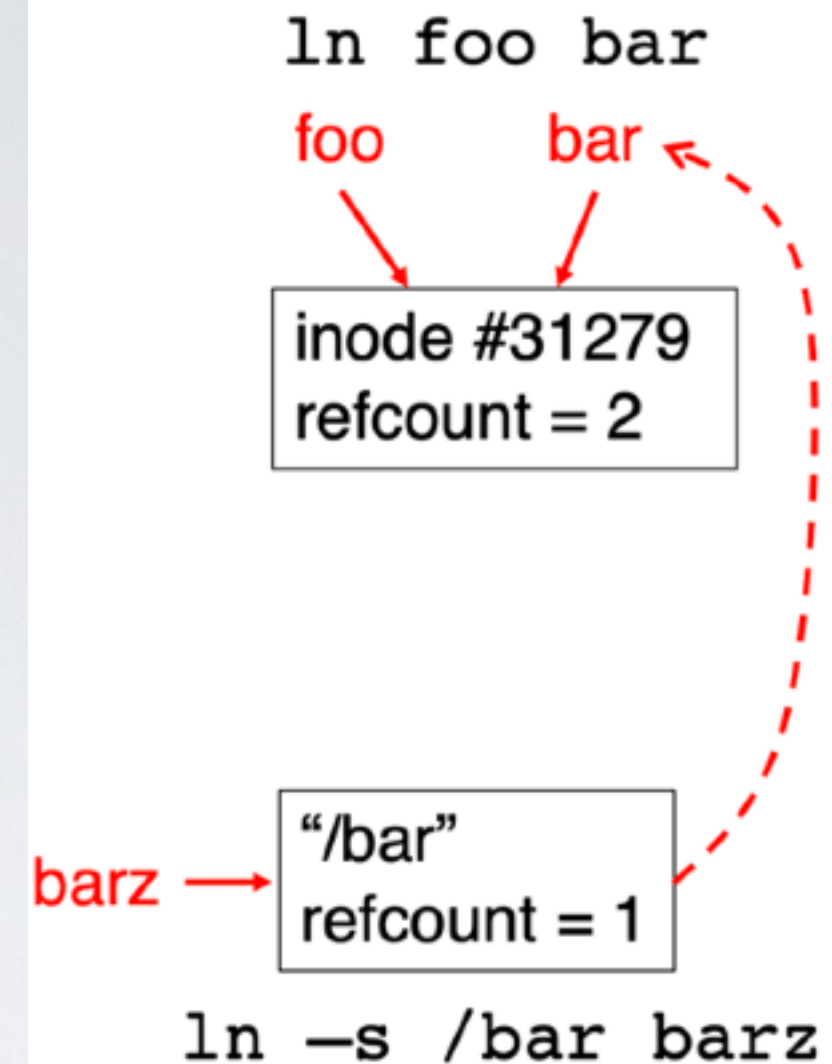
# Hard and Soft Links (synonyms)

## More than one dir entry can refer to a given file

- Hard link creates a synonym for file
- Unix stores count of pointers ("hard links") to inode
- If one of the links is removed (e.g., rm), the data are still accessible through any other link that remains
- If all links are removed, the space occupied by the data is freed

## Soft symbolic links = synonyms for names

- Point to a file/dir name, but object can be deleted from underneath it (or never exist)
- Unix implements like directories: inode has special "symlink" bit set and contains name of link target
- When the file system encounters a soft link it automatically translates it (if possible).



# Improving Reliability

with Log-Structured File system (LFS)  
and Journaling File System (ext3)